



Office Order No. 80

Dated 24/03/2026

Sub. Methodology to be adopted for installation of the smart meters for Distribution Transformers.

The deployment of Smart Metering for Distribution Transformers under the RDSS project is currently in progress. The designated AMISPs—Polaris (Hooghly Smart Metering), GVPR, and Apraava —are actively installing meters across various sites within the jurisdiction of WBSEDCL.

In this context, the methodology for installation was discussed and intimated to site offices as well as instructed to the AMISPs vide MoM: CE/Comm/Tender/RDSS/Smart Meter/AMISP/Com8/2133 dated 11.07.2025 and through a VC dated 08.09.2025. However, field-level observations during the implementation phase have necessitated a revision of approach regarding utilization of the existing cable. Additionally, other minor technical issues have surfaced that require immediate rectification. To ensure technical uniformity and address these challenges, the following methodology shall be strictly adopted for all future DT smart meter installations.

1. Cable Issues

- I. For existing installed cables (for single/3½/4 cores), the cable shall not be replaced unless the cable is in damaged condition.
- II. The existing cable between the Transformer LT Bushing and the LT Kiosk may be utilized for the connection between the DTR Bushing and the incoming of Metering Box. The remaining portion, from the outgoing of Metering Box to the LT Kiosk, must be completed using a new cable (**Approach -1**).
- III. **Approach -1 method will also be effective for the case of short of length of existing LT cable.**
- IV. The existing cable may be bisected and two ends of one segment of the bisected cable shall be utilized for connecting between LT bushing and Incoming of Metering arrangement. Another segment of the bisected cable shall be used for connection between Outgoing of Metering arrangement and LT kiosk termination points (**Approach – 2**).
- V. **Thus, the earlier approach of replacing the installed existing LT cables shall now stand cancelled and either one of the below-mentioned approaches is to be adopted for future installations.**

Illustrations are attached as reference for installation of the smart meter by using existing cables as per point no. II, III & IV)

2. Existing dismantled meter, CT & Cable (For damaged existing cable) Return Issues:-

- The Station-Manager(s) shall create a Reservation Number for return for the lot of the non-smart meters in SAP-ERP under departmental return head only after the validation of the submitted data sheet by the AMISP(s).
- Upon submission of the Reservation Number to the store, the store shall generate and post the SRN.
- This SRN shall be authenticated by the Store-in-Charge and AMISP shall upload the same in their WFM.

WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED

(A GOVERNMENT OF WEST BENGAL ENTERPRISE)



Office of The Director: Distribution

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- A dedicated store location shall be created in SAP-ERP for accepting the return.

The above methodology shall need to be followed scrupulously while installation of the smart meters in Distribution Transformers.

3. Additional required materials to be procured by WBSEDCL for Approach-1:-

- 3 – 4 meter long Heat shrinkable LT sleeves (for 95 sq. MM 4 core XLPE cable - 19/6 or 25/8 Medium-Wall Heat Shrink. (Thickness 2.5 MM) & for 185 Sq. MM 4 core XLPE cable –36/12 Medium-wall Heat shrink (thickness 2.5 MM).
- One LT breakouts (Size 55/25 for 95 Sq. MM cable, Size 68/25 for 185 sq. MM cable).

4. Additional required materials to be procured by WBSEDCL for Approach-2 :

- 6 – 8-meter-long Heat shrinkable LT sleeves (for 95 sq. MM 4 core XLPE cable - 19/6 or 25/8 Medium-Wall Heat Shrink. (Thickness 2.5 MM) & for 185 Sq. MM 4 core XLPE cable –36/12 Medium-wall Heat shrink (thickness 2.5 MM).
- Two nos LT breakouts (Size 55/25 for 95 Sq. MM cable, Size 68/25 for 185 sq.MM cable).

All concerned must adhere to these guidelines without deviation and all controlling officers, are hereby, instructed to follow the instructions scrupulously and transmit the message downstream for compliance.


(S. Mukherjee)

Director (Distribution) : WBSEDCL

Memo No.DD/OO/173

Dated 23/03/2026

Distribution :-

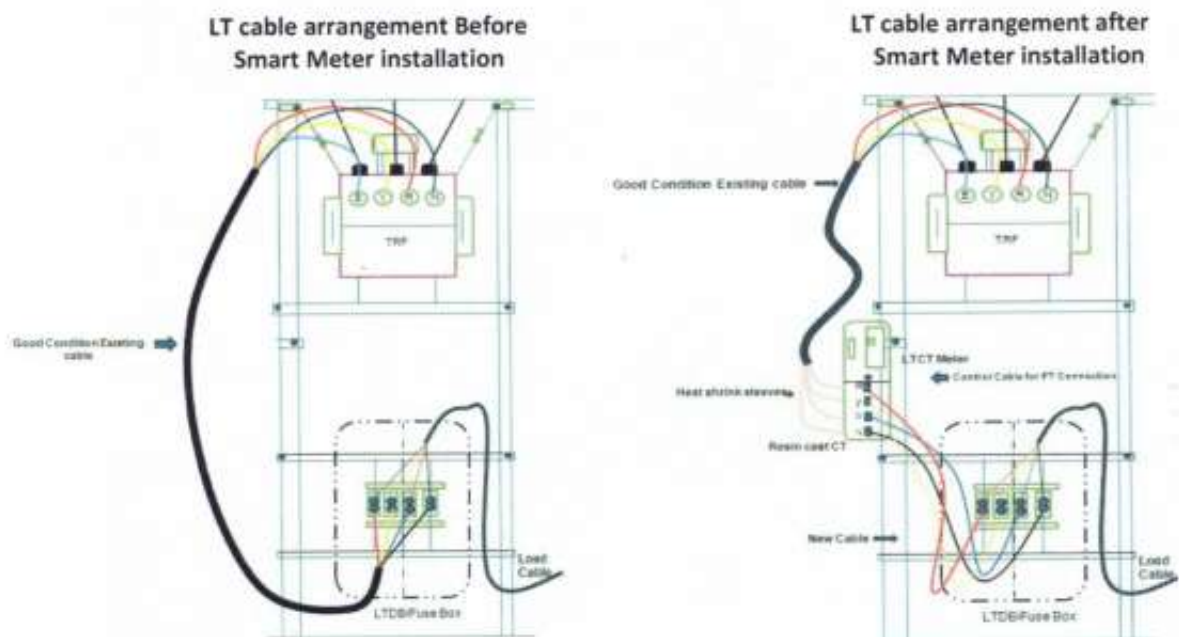
1. The Director (Project), WBSEDCL.
2. The Executive Director (Distribution/Project), WBSEDCL.
3. The Chief Engineer, Dist-South/Dist-North/Communication/REM/DTD/P&E/Project-II, WBSEDCL.
4. All Zonal Managers, WBSEDCL.
5. All Regional Managers, WBSEDCL.
6. All Divisional Managers, WBSEDCL.
7. The Sr. P.S. to CMD, WBSEDCL.



Approach of Metering:

Approach – 1:

- **Bus-bar Installation:** Four insulated bus-bars(25 mm width X 8 mm thickness for 95 sq. MM Cable, 40mm width X 10 mm thickness for 185 sq. MM cable)shall be securely mounted within the designated chamber of the SMC enclosure/box, ensuring adequate clearance and structural stability.
- **CT Integration:** The Current Transformers (CTs) must be slid over the respective bus-bars before final fixing to ensure accurate current sensing for each phase and the neutral.
- **Cable Preparation:** The existing cable shall be disconnected from the LT Kiosk and redirected to the incoming side of the Metering Unit, where it will be secured using appropriate lugs and bolts. A new single-core cable of the required cross-section shall be utilized to bridge the connection between the outgoing side of the Metering Unit and the LT Kiosk.
- **Core Insulation:** LT Heat-Shrinkable sleeves must be applied to the termination point of the existing cable. A heat gun shall be used to apply uniform heat, ensuring the sleeves shrink tightly to provide a robust, moisture-proof seal over the cores.
- **Environmental Sealing:** A 4-way LT Cable Breakout (boot) must be installed at the entry point of the existing cable into the Metering Unit to hermetically seal the junction between the individual cores and the main cable sheath.
- **Termination:** High-quality Compression Lugs (total 8 units) must be professionally crimped onto the four cores of both cable sides.
- **Final Connection:** The cable lugs shall be securely fastened onto the incoming and outgoing sides of the respective bus-bars using bolts and washers, ensuring high conductivity and a vibration-resistant joint.



Amrit Singh

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Approach – 2:

- **Bus-bar Installation:** Four insulated bus-bars(25 mm width X 8 mm thickness for 95 sq. MM Cable, 40mm width X 10 mm thickness for 185 sq. MM cable)shall be securely mounted within the designated chamber of the SMC enclosure/box, ensuring adequate clearance and structural stability.
- **CT Integration:** The Current Transformers (CTs) must be slid over the respective bus-bars before final fixing to ensure accurate current sensing for each phase and the neutral.
- **Cable Preparation:** The incoming/outgoing LT cable shall be precisely bisected at a suitable position to facilitate the new connection layout.
- **Core Insulation:** LT Heat-Shrinkable sleeves must be applied to each of the four cores on both separated cable ends. A heat gun shall be used to apply uniform heat, ensuring the sleeves shrink tightly to provide a robust, moisture-proof seal over the cores.
- **Environmental Sealing:** 4-way LT Cable Breakouts (boots) must be installed at each cable end to hermetically seal the junction between the individual cores and the main cable sheath.
- **Termination:** High-quality Compression Lugs (total 8 units) must be professionally crimped onto the four cores of both cable sides.
- **Final Connection:** The cable lugs shall be securely fastened onto the incoming and outgoing sides of the respective bus-bars using bolts and washers, ensuring high conductivity and a vibration-resistant joint.



Signature
24/03/22